Pacific Graphics (2026)
Y. He, N. Thürey, and L. Liu (Editors)



# A Lightweight Image-Space Deformation Method

1023

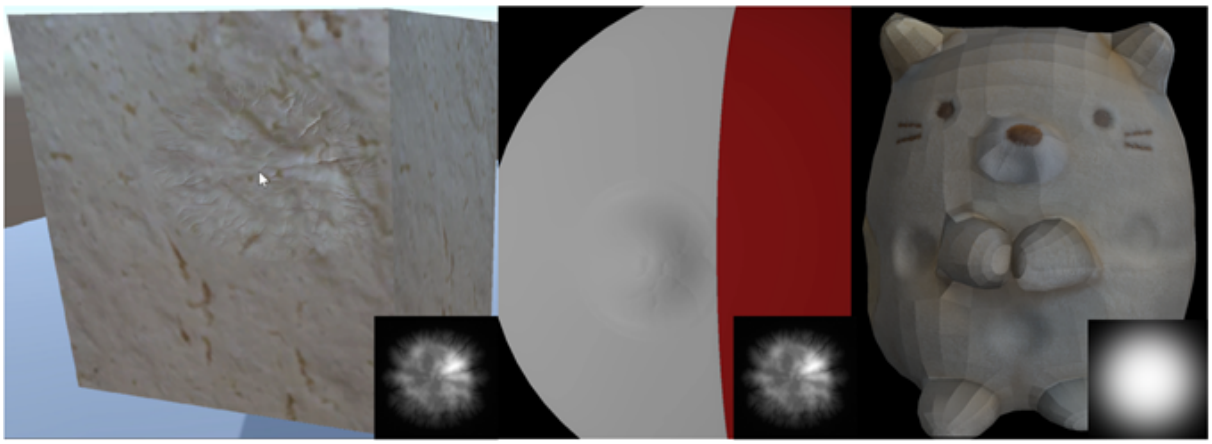


**Figure 1:** *Results of our image-space deformation method. The height map shown at the bottom right provides direct control over fine-grained, subtle surface deformations.*

**Abstract**

*We present a lightweight screen-space method for generating subtle deformation effects. Conventional parallax mapping uses a height map to add apparent geometric detail and surface relief without increasing mesh complexity. In contrast, our method re-purposes height-guided texture-coordinate perturbation to produce time-varying deformation. An input height map controls the spatial distribution and magnitude of displacement, while a 2D animation signal defines its motion. The method can be implemented entirely in a post-processing shader, requiring no vertex updates, skeletal rigging, or physical simulation. Its cost is therefore largely independent of mesh complexity, making it suitable for real-time and resource-constrained applications. Height maps may be estimated automatically, generated with computer-vision assistance, or manually authored for direct control.*

**CCS Concepts**

• ***Computing methodologies*** → ***Procedural animation;***

## 1. Introduction

Subtle motions such as breathing, pulsation, and soft bulging are commonly implemented by deforming a mesh or driving a rig. For effects with limited silhouette change, this introduces geometric and authoring overhead that may be unnecessary. Image deformation provides a lighter alternative; moving-least-squares warping, for example, constructs smooth 2D deformations from user controls [SMW06].

Texture-based rendering offers a related precedent for avoiding dense geometry. Bump mapping modifies surface appearance without explicitly modeling fine geometry [Bli78]. Parallax mapping instead perturbs texture coordinates according to a height map and viewing direction, adding apparent relief to a simple surface [KTI*01]. Relief mapping improves depth and occlusion cues through additional per-pixel sampling [POC05].

Our main contribution is a simple, lightweight deformation framework that repurposes parallax-style texture perturbation as an animation operator. Instead of using a height map only to add apparent surface detail, the proposed method combines it with a time-varying 2D control field to generate rich, fine-grained, and subtle deformation directly in the rendered image. The effect is implemented as a screen-space post-processing shader, requiring neither vertex updates nor skeletal rigging. Height maps can be created au-

© 2026 The Author(s).
Proceedings published by Eurographics - The European Association for Computer Graphics.
This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

tomatically using a MediaPipe-based pipeline, semi-automatically through image-assisted measurement, or manually for direct artistic control. This flexible workflow makes the method easy to integrate into different content-creation pipelines, while remaining well suited to small-amplitude motions that do not require major visibility, silhouette, or topology changes.

## 2. Method

### 2.1. Inputs

The method takes a rendered source image $I_0$, a renderer-ready UV-space height map $H$, and a time-varying 2D control field $\mathbf{c}(\mathbf{u},t)$. As shown in Figure 2, The height map can be generated through two pipelines. In the semi-auto measurement-based pipeline, ruler-calibrated neutral and pressed images are compared in Fiji/ImageJ to obtain physical displacements, which are interpolated into a smooth local deformation patch. In the learning-based pipeline, MediaPipe tracks visible cheek landmarks during face-press recordings, and a trained model predicts a dense regional deformation map. Both pipelines produce a $1024 \times 1024$ grayscale patch positioned in the corresponding region of the face UV layout. A value of 128 represents the neutral surface, while darker values encode stronger inward deformation. Pixels outside the local patch remain neutral or are excluded using a mask.

### 2.2. Height-guided warp

For normalized screen coordinate $\mathbf{u}$, we compute

$$\begin{aligned} \Delta\mathbf{u}(\mathbf{u},t) &= \alpha\, w(H(\mathbf{u}))\, \mathbf{c}(\mathbf{u},t), \\ I_t(\mathbf{u}) &= I_0(\mathbf{u} - \Delta\mathbf{u}(\mathbf{u},t)), \end{aligned} \tag{1}$$

where $\alpha$ controls amplitude and $w(\cdot)$ optionally remaps the height response. A sinusoidal global field produces oscillation, while radial, keyframed, or artist-authored fields produce localized expansion, compression, and directional motion. In contrast to conventional parallax mapping, $\mathbf{c}$ is an animation signal rather than a function of viewing direction; the height map therefore acts as a deformation profile.

### 2.3. Shader implementation

Equation (1) is evaluated in a full-screen fragment shader using a decal-based rendering strategy. The global height map provides the base deformation profile, while localized height-map patches are applied as decals at selected surface regions, each with its own position, scale, and falloff radius. These decals are composited in UV or screen space before the final displacement is computed, allowing local deformation details to be added without modifying the underlying texture or geometry. The procedural-control variant performs the required height and decal lookups, computes the combined offset, and samples the source image for color and alpha. The pass does not update vertices, evaluate a rig, or ray-march the height field; its runtime therefore depends mainly on the processed image area and the number of active decals rather than mesh complexity. Boundary samples are clamped, padded, or faded to reduce edge artifacts.

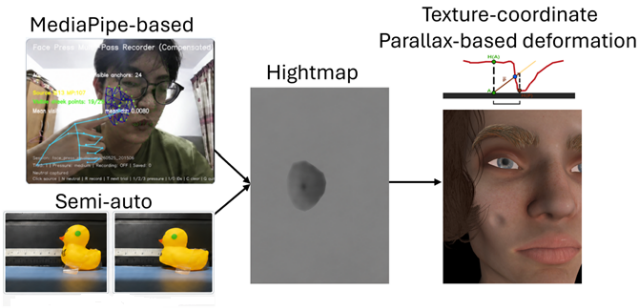



**Figure 2:** *Method overview.*

## 3. Results

Figure 1 presents number of assets examples driven by control signals. The method produces smooth, localized deformation while retaining the original texture appearance for small offsets. At output resolution on GPU/device, the post-processing pass requires GPU time in ms per frame and achieves FPS; the overhead over displaying the undeformed image is percentage. A vertex-deformation baseline with vertex count vertices requires GPU time in ms, corresponding to relative speedup or reduction.

The method is most effective for low-amplitude motion. Large offsets may expose unavailable samples, stretch features, or produce incorrect silhouettes because disocclusion and true geometric visibility are not reconstructed.

## 4. Conclusion

We presented a lightweight image-space deformation method that reinterprets parallax-style texture perturbation as an animation operator. A height map modulates a 2D control field, and a single post-processing shader generates subtle deformation without modifying geometry or constructing a rig. Future work will investigate occlusion-aware sampling and improved boundary handling for larger motions.

## References

[Bli78] BLINN J. F.: Simulation of wrinkled surfaces. In *Proceedings of the 5th Annual Conference on Computer Graphics and Interactive Techniques* (1978), ACM, pp. 286–292. doi:10.1145/800248.507101. 1

[KTI*01] KANEKO T., TAKAHEI T., INAMI M., KAWAKAMI N., YANAGIDA Y., MAEDA T., TACHI S.: Detailed shape representation with parallax mapping. In *Proceedings of the International Conference on Artificial Reality and Telexistence (ICAT)* (2001), pp. 205–208. 1

[POC05] POLICARPO F., OLIVEIRA M. M., COMBA J. L. D.: Real-time relief mapping on arbitrary polygonal surfaces. In *Proceedings of the 2005 Symposium on Interactive 3D Graphics and Games* (2005), ACM, pp. 155–162. doi:10.1145/1053427.1053453. 1

[SMW06] SCHAEFER S., MCPHAIL T., WARREN J.: Image deformation using moving least squares. *ACM Transactions on Graphics 25*, 3 (2006), 533–540. doi:10.1145/1141911.1141920. 1

© 2026 The Author(s).
Proceedings published by Eurographics - The European Association for Computer Graphics.